VALENTINA ZANGIROLAMI

valentina.zangirolami@gmail.com | Website | ValentinaZangirolami

Research Interests Statistical reinforcement learning, Deep learning, Statistical learning, Foundation models.

Research Network Member of Prof. **Chengchun Shi**'s research group, contributing to research discussions on reinforcement learning, large language models, and reinforcement learning from human feedback.

Work Experience

Research Fellow November 2026 – ongoing

Institute for Applied Mathematics and Information Technologies "Enrico Magenes" (CNR-IMATI) *Milan, Italy*

- Research in statistics within the project *Near-Earth Asteroids: a Benchmark model for Exploration and Science* (NEA-BES), conducted by CNR-IMATI in partnership with the Italian National Institute for Astrophysics (INAF).
- Apply statistical methodology within an interdisciplinary collaboration between applied mathematics and astrophysics, under the scientific supervision of Dr. Elisa Maria Alessi.

Statistical Analyst November 2026 – ongoing

University of Milano-Bicocca *Milan, Italy*

- Conduct statistical analyses of questionnaires completed by students and author analytical reports for the Directorate of Teaching, Student Services and Internationalisation, within a project led by Prof. Sonia Migliorati.

Data Science Analyst July 2022 – Oct 2022

Accenture (Media & Communications – Telecommunications) *Milan, Italy*

- Conducted data quality assessment, data analysis, feature selection, and feature engineering.
- Worked with AWS S3 and EMR, DBeaver/Apache Hive, and PySpark.

Data Engineering Analyst Jan 2022 – June 2022

Accenture (Banking & Finance – ESG) *Milan, Italy*

- Managament of Engineering projects with definition of milestones and coordinating all the project's processes.
- Gained experience in business model strategy and business process design.

Education

Visiting PhD Student in Statistics Feb 2025 – July 2026

London School of Economics and Political Science *London, UK*

- Research interests: *Statistical inference for reinforcement learning with temporally correlated rewards.*
- **Supervisor**: Prof. Chengchun Shi

PhD in Statistics Oct 2022 – Oct 2026

University of Milano-Bicocca *Milan, Italy*

- Thesis: *On statistical methods for reinforcement learning with complex data.*
- **Supervisor**: Prof. Chengchun Shi **Co-supervisor**: Prof. Matteo Borrotti
- Expected graduation at February/March 2027.

Master's Degree in Statistical and Economic Sciences Oct 2019 – Jan 2022

University of Milano-Bicocca *Milan, Italy*

- Thesis: *Recurrent structures of deep reinforcement learning for self-driving cars*.
- **Supervisor**: Prof. Matteo Borrotti **Co-supervisor**: Prof. Antonio Candelieri

- Grade: 110/110 with honors

Bachelor's Degree in Statistical and Economic Sciences Oct 2016 – July 2019
University of Milano-Bicocca *Milan, Italy*

- Thesis: *The mobile telecommunications market: the entry of Iliad*.
- **Supervisor**: Dott. Christian Garavaglia

Publications

Refereed Journal Articles

- **Zangirolami, V.** and Borrotti, M. (2024). *Dealing with uncertainty: Balancing exploration and exploitation in deep recurrent reinforcement learning*. *Knowledge-Based Systems*, 293.

Working Papers

- Ye, K., Gong, S., Zhou, H., **Zangirolami, V.***, and Shi, C.* (2026+). *Can Tabular Foundation Models Amortize Statistical Inference?*. arXiv preprint arXiv:2609.33114. (*corresponding authors)

Book Chapters and Proceedings

- **Zangirolami, V.**, Pavesi, F., and Zanotti, M. (2026). *Safe exogenous-state reinforcement learning for water tank systems*. *Statistical Science: From Theory to Applied Research IV*.
- Pedrazzini, E., **Zangirolami, V.**, Migliorati, S., and Borrotti, M. (2026). *Beta-Distributed Proximal Policy Optimization for Autonomous Driving with Grad-CAM Explainability*. *Artificial Intelligence Applications and Innovations*.
- Pedrazzini, E., **Zangirolami, V.**, Migliorati, S., and Borrotti, M. (2026). *Flexible Beta: A novel policy distribution for proximal policy optimization*. *Statistical Science: From Theory to Applied Research IV*.
- Zanotti, M., **Zangirolami, V.**, and Pavesi, F. (2026). *An evaluation of ensemble strategies for time-series anomaly detection*. *Statistical Science: From Theory to Applied Research IV*.
- Pavesi, F., Zanotti, M., and **Zangirolami, V.** (2026). *A characterization of Gaussian processes over flat tori*. *Statistical Science: From Theory to Applied Research IV*.

Invited Talks

- **Zangirolami, V.**, Borrotti, M., and Candelieri, A. *Conformal prediction for safe decision-making*. **HiTEc Meeting & Workshop on Complex Data in Econometrics and Statistics**, Limassol, Cyprus, July 8–9, 2025. **Session chair**.
- **Zangirolami, V.**, Borrotti, M., and Candelieri, A. *Using conformal prediction for Gaussian process regression in dynamic systems*. **Statalk 2025**, Milan, Italy, June 13–14, 2025.
- **Zangirolami, V.** *Enhancing data efficiency in online deep reinforcement learning under partial observability*. **18th International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics)**, London, UK, Dec 2024.

Contributed Talks

- **Zangirolami, V.**, Pavesi, F., and Zanotti, M. (2026). *Safe exogenous-state reinforcement learning for water tank systems*. **Joint Meeting SIS-FENStatS 2026**, Rome, Italy, June 22–25, 2026.

Organization of Scientific Events

- Serving as a **Program Chair** for the NeurIPS 2026 Workshop *Foundations of LLM Post-Training in Changing Environments*, mainly organized by Dr. F. Quinzan, Dr. G. Peyré, Prof. P. Rebeschini, Prof. C. Shi, Prof. F. Liu.

Grants

AIRR Gateway Project Sept 2026 - Dec 2026

- Member of the research team for the grant project *AIRR Gateway*, entitled *"Does Reasoning Drive LLM Decisions? A Causal Evaluation"*. PI: Prof. Chengchun Shi, Co-PI: Prof. Giulia Livieri

AIRR Gateway Project May 2026 - Aug 2026

- Member of the research team for the grant project *AIRR Gateway*, entitled *"Reasoning-Enhanced Multimodal AI-Generated Content Detection"*. PI: Prof. Giulia Livieri, Co-PI: Prof. Chengchun Shi

AIRR Gateway Project Feb 2026 - May 2026

- Member of the research team for the grant project *AIRR Gateway*, entitled *"Lowering the Computational Barrier: Resource- and Energy-Efficient Reinforcement Learning for LLM Reasoning"*. PI: Prof. Chengchun Shi, Co-PI: Prof. Giulia Livieri

Research Awards

Workshops (Selected/Invited Participation)

Sanya Workshop on Causality and Machine Learning Jan 12–16, 2026
Tsinghua University – Tsinghua Sanya International Mathematics Forum (TSIMF) *Sanya, China*

- Invited participant.

The Data Research Camp July 15–18, 2024
Department of Statistics, Computer Science, Applications, University of Florence *Florence, Italy*

- Selected participant in a workshop involving group activities focused on developing new models for clinical and protein data.
- Received a travel award covering accommodation.

Data Competitions

Stats Under the Stars 6 June 20–21, 2023
Italian Statistical Society (SIS) *Ancona, Italy*

- Awarded second place for Best Objective Prediction.
- Challenge topic: estimation of portfolio weights for financial assets using real data provided by Flow Traders.

Lab Data Challenges: Electricity Load Forecast Challenge May 2021
University of Milano-Bicocca *Milan, Italy*

- Awarded first place for Best Prediction.
- Challenge topic: long-term electricity load forecasting using real data provided by Repower.

Referee service

- Journal of Statistical Computation and Simulation

Thesis Supervision

- **Co-supervisor** of master's students working on reinforcement learning and conformal prediction topics, including: Giulia De Innocentiis (*Conformal Prediction for Neural Fitted Q-iteration*), Giorgia Faccaroni (*Conformal Prediction for Fitted Q-iteration*), Antonio Monforte (*Proximal Policy Optimization for Autonomous Driving*), and Roberto Magnacavallo (*Conformal Prediction for Superconductivity Data*).

Teaching Experience

Teaching Assistant AY 2025 – 2026, 2026 – 2027
Statistics course, Bachelor's Degree *Bocconi University, Italy*

- Supported teaching activities for the undergraduate statistics course in Italian and English.

Tutor, Machine Learning and AI Course Dec 2025 – May 2026
Scientific Degree Plan (PLS) *University of Milano-Bicocca, Italy*

- Organized and delivered a machine learning and artificial intelligence crash course which includes also practical session testing AI models using Hugging Face.
- Language of instruction: Italian.

Tutor Dec 2025 – May 2026
Scientific Degree Plan (PLS) *University of Milano-Bicocca, Italy*

- Prepared materials for laboratory activities and supported student projects on classification models in R.
- Language of instruction: Italian.

Adjunct Lecturer, Business Statistics AY 2024 – 2025 / 2025 – 2026
Bachelor's Degree in Economics and Management *University of Milan, Italy*

- Delivered lectures in Italian using R programming, application of linear models, decision trees, clustering, and time series analysis.
- Language of instruction: Italian.
- Course materials available at https://github.com/ValentinaZangirolami/Statistica_decisioni_aziendali.

Adjunct Lecturer, Statistical Learning AY 2024 – 2025
Master's Degree in Artificial Intelligence for Science and Technology *University of Milano-Bicocca, Italy*

- Delivered lectures in English on statistical learning methods in R, including practical exercises on decision trees.
- Language of instruction: English.
- Course materials available at https://github.com/ValentinaZangirolami/Statistical-Learning.

Tutor Sep 2024
Stempathy Project *IISS "M.K. GANDHI", Italy*

- Prepared materials for laboratory activities and led lab sessions on regression models in R.
- Language of instruction: Italian.

Tutor June 2024 – July 2024
Summer School PNRR: Laboratory of Statistics *University of Milano-Bicocca, Italy*

- Contribution to the statistics laboratory activities, supporting students in data analysis of the mosquito distribution globally.
- Language of instruction: Italian.

Tutor Feb 2024 – May 2024
Scientific Degree Plan (PLS) *University of Milano-Bicocca, Italy*

- Prepared materials for laboratory activities and supported student projects on regression models in R.
- Language of instruction: Italian.

Adjunct Lecturer, Statistical Modelling AY 2023 – 2024

Bachelor's Degree in Artificial Intelligence *University of Milano-Bicocca, Italy*

- Delivered practical sessions in English on simple linear regression, multiple linear regression, and generalized linear models, with applications in R, and written exercises on hypothesis testing.
- Language of instruction: English.

Tutor, Statistical Learning AY 2022-2023 / 2023 – 2024

Master's Degree in Artificial Intelligence for Science and Technology *University of Milano-Bicocca, Italy*

- Delivered tutorials in English using R programming and tidyverse workflows.
- Introduced GitHub-based project workflows; example final project: data analysis and statistical learning models for churn prediction.
- Led a mock final exam reviewing key theoretical topics in statistical learning.
- Language of instruction: English.

Other Courses (Extra-Curricular Professional Courses)

Natural Language Processing Specialization Sep 2022

DeepLearning.AI (Coursera)

- Certification available upon request.

AI for Medicine Specialization Sep 2022

DeepLearning.AI (Coursera)

- Certification available upon request.

Data Engineering and Machine Learning Using Spark Aug 2022

IBM (Coursera)

- Certification available upon request.

Skills

Programming Languages: R (advanced), Python (advanced), SQL (intermediate), C/C++ (basic)
Markup Languages: LaTeX (advanced), HTML (basic)
Data Visualization: Power BI, Tableau
Other Statistical Software: KNIME, Gretl
Developer Tools: Jupyter Notebooks, Git, AWS (S3 and EMR), IBM Cloud (Watson Machine Learning)
Operating Systems: Windows, macOS, Linux (basic)
Python Libraries: NumPy, pandas, scikit-learn, TensorFlow, PyTorch, Keras, Matplotlib, Seaborn, OpenCV, PySpark, OpenAI Gym
Languages: Italian (native), English (fluent)

November 2026

Valentina Zangirolami